MÓDULO 2-ALGORITMOS EN R*

Cañadas Ian[1], Bresci, Joaquin[2], Isgro, Mateo[1], La Rocca, Nicolás[1], Salvo, Luca[1], and Zanetti, Bautista[1]

[1] Universidad Nacional de Cuyo, Instituto de Ingenería Industrial,POBOX 5502,C.P.:5500 Mendoza , Argentina
2

lncs@uncuyo.com
http://www.uncuyo.edu.ar/contacto
3

Abstract. R Posit en un software online que permite utilizar lenguaje R y trabajar tambien con matrices de forma similar a Mathlab

Keywords: Lenguaje R[1–3] · TYHM · Algoritmos

```
knitr::opts_chunk$set(echo = TRUE,
                      out.extra = "")  # This forces knitr to label all figures.
```

```
knitr::opts_chunk$set(echo = TRUE)
# Definición de variables y operaciones básicas
a <- 40
b <- 60
c <- b - a
print(c)
```

```
## [1] 20
```

```
knitr::opts_chunk$set(echo = TRUE)
# Resumen de base de datos interna
summary(cars)
```

```
##      speed           dist
##  Min.   : 4.0   Min.   :  2.00
##  1st Qu.:12.0   1st Qu.: 26.00
##  Median :15.0   Median : 36.00
##  Mean   :15.4   Mean   : 42.98
##  3rd Qu.:19.0   3rd Qu.: 56.00
##  Max.   :25.0   Max.   :120.00
```

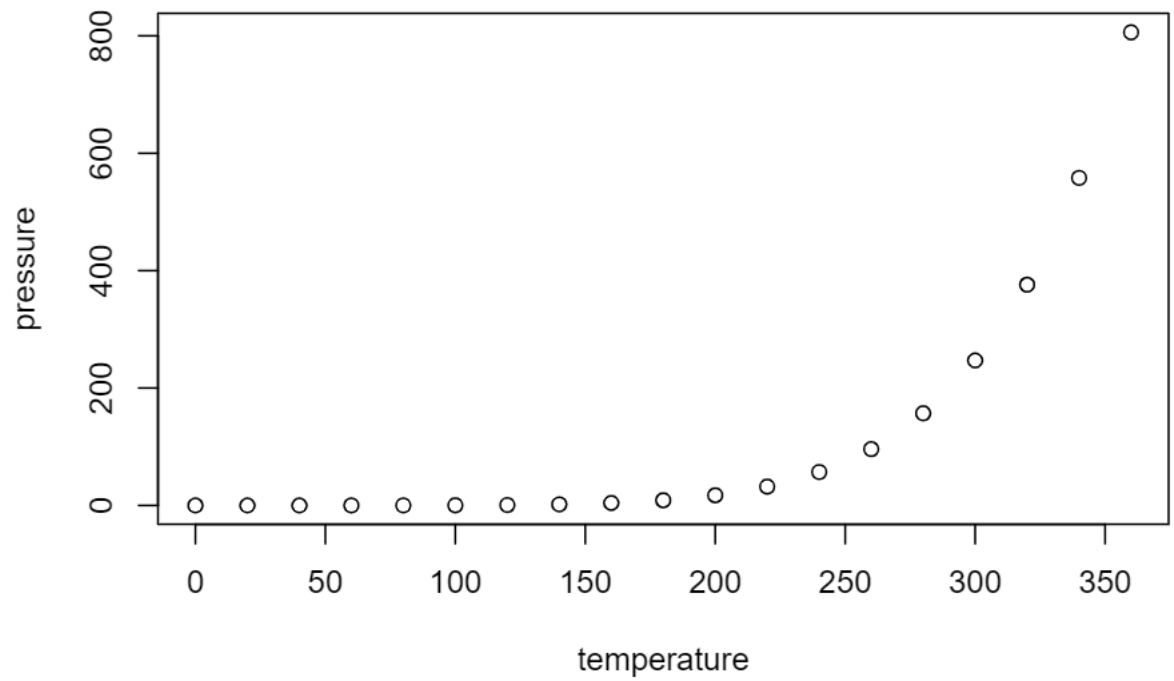
* Supported by organization x.

```r
print(cars)
```

```
##    speed dist
## 1      4    2
## 2      4   10
## 3      7    4
## 4      7   22
## 5      8   16
## 6      9   10
## 7     10   18
## 8     10   26
## 9     10   34
## 10    11   17
## 11    11   28
## 12    12   14
## 13    12   20
## 14    12   24
## 15    12   28
## 16    13   26
## 17    13   34
## 18    13   34
## 19    13   46
## 20    14   26
## 21    14   36
## 22    14   60
## 23    14   80
## 24    15   20
## 25    15   26
## 26    15   54
## 27    16   32
## 28    16   40
## 29    17   32
## 30    17   40
## 31    17   50
## 32    18   42
## 33    18   56
## 34    18   76
## 35    18   84
## 36    19   36
## 37    19   46
## 38    19   68
## 39    20   32
## 40    20   48
## 41    20   52
## 42    20   56
```

```
## 43     20    64
## 44     22    66
## 45     23    54
## 46     24    70
## 47     24    92
## 48     24    93
## 49     24   120
## 50     25    85
```

```r
plot(pressure)
```

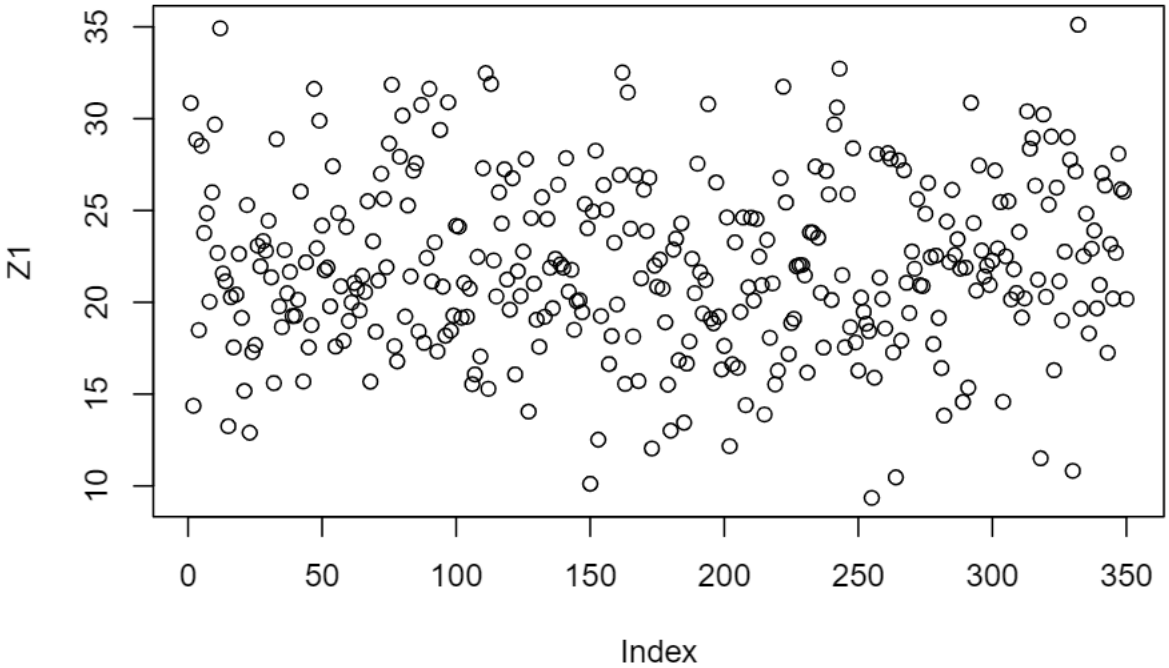



```r
knitr::opts_chunk$set(echo = TRUE)
 # Generar 350 datos con media 22 y desviación estándar 5
Z1 <- rnorm(350, 22, 5)
print(Z1)
```

```
##   [1] 30.85251 14.36210 28.85538 18.48785 28.52552 23.77326 24.85176 20.03174
##   [9] 25.98958 29.68303 22.68456 34.92328 21.57268 21.14118 13.25227 20.27875
##  [17] 17.54582 20.41768 22.63839 19.15144 15.17479 25.29595 12.90169 17.29010
##  [25] 17.67862 23.07762 21.95935 23.33564 22.80481 24.44091 21.35878 15.60152
##  [33] 28.88478 19.77589 18.64571 22.84153 20.49139 21.65751 19.25376 19.25967
##  [41] 20.13897 26.03539 15.68754 22.17755 17.54850 18.75386 31.62360 22.94292
##  [49] 29.88788 24.18193 21.74104 21.89112 19.77965 27.41040 17.58867 24.85744
```

```
##  [57] 20.87420 17.89714 24.10550 18.98553 19.99386 21.05494 20.73557 19.55421
##  [65] 21.43695 20.56454 25.50437 15.67936 23.32092 18.39873 21.18676 27.00343
##  [73] 25.62938 21.91125 28.64615 31.85437 17.61453 16.78294 27.92999 30.17206
##  [81] 19.21978 25.27166 21.41632 27.16664 27.57862 18.41229 30.74964 17.78892
##  [89] 22.41493 31.62380 21.13342 23.26173 17.32537 29.38702 20.85658 18.18258
##  [97] 30.89150 18.45627 19.27918 24.17217 24.11595 19.15640 21.05520 19.21044
## [105] 20.74523 15.54142 16.07589 22.47254 17.05321 27.30202 32.47962 15.29042
## [113] 31.89329 22.28172 20.32192 25.99113 24.29050 27.24426 21.23867 19.59824
## [121] 26.75824 16.07601 21.69524 20.34049 22.76153 27.79034 14.04782 24.59008
## [129] 21.01498 19.05251 17.57829 25.72405 19.21405 24.53864 21.88575 19.67774
## [137] 22.36669 26.40321 22.07016 21.88538 27.84908 20.57920 21.77450 18.50192
## [145] 20.05700 20.12294 19.46250 25.35163 24.03487 10.11831 24.94677 28.25591
## [153] 12.51965 19.25114 26.39130 25.04114 16.63780 18.16349 23.24480 19.88654
## [161] 26.93434 32.51480 15.55672 31.42987 24.01200 18.13990 26.92128 15.71295
## [169] 21.30627 26.12761 23.87921 26.78459 12.04137 21.99890 20.84498 22.32537
## [177] 20.73588 18.90936 15.51399 13.01073 22.87128 23.46738 16.84380 24.29927
## [185] 13.44452 16.66276 17.87016 22.36820 20.50185 27.54866 21.64036 19.38956
## [193] 21.21364 30.79439 19.09967 18.85625 26.52350 19.21889 16.34617 17.62318
## [201] 24.62418 12.16667 16.63010 23.26533 16.43359 19.47721 24.61620 14.39768
## [209] 20.81981 24.60896 20.09193 24.53086 22.48835 20.92848 13.89204 23.39713
## [217] 18.07071 21.02943 15.52909 16.27917 26.77632 31.73787 25.43587 17.18415
## [225] 18.86944 19.10429 21.96106 22.02651 22.02407 21.46609 16.17336 23.80992
## [233] 23.80186 27.39121 23.52167 20.52252 17.53397 27.15106 25.87489 20.12376
## [241] 29.69046 30.60889 32.72468 21.47939 17.54991 25.89273 18.65052 28.38852
## [249] 17.81560 16.27696 20.25512 19.48587 18.83061 18.42419  9.35229 15.89359
## [257] 28.07790 21.34296 20.18132 18.57860 28.11946 27.81219 17.26789 10.46779
## [265] 27.72920 17.91254 27.19046 21.05742 19.41530 22.76624 21.82651 25.61441
## [273] 20.93234 20.88513 24.82008 26.50120 22.45466 17.72845 22.54001 19.14862
## [281] 16.41913 13.83104 24.39675 22.18986 26.12175 22.58011 23.43667 21.83750
## [289] 14.57542 21.88883 15.35225 30.86536 24.31259 20.63635 27.45203 22.81499
## [297] 21.41548 21.99540 20.94621 22.26949 27.17425 22.93516 25.45488 14.58138
## [305] 22.49633 25.50328 20.15590 21.79938 20.50829 23.83522 19.16997 20.21123
## [313] 30.39920 28.36758 28.93878 26.35032 21.23256 11.50419 30.23079 20.29676
## [321] 25.32067 29.02904 16.30026 26.24508 21.14480 19.01442 22.75966 28.99086
## [329] 27.76777 10.82283 27.12637 35.11562 19.65608 22.51545 24.82488 18.30736
## [337] 22.91852 23.90411 19.67073 20.94917 27.02933 26.36273 17.25361 23.16673
## [345] 20.20358 22.69860 28.08881 26.15297 26.02121 20.17361
```

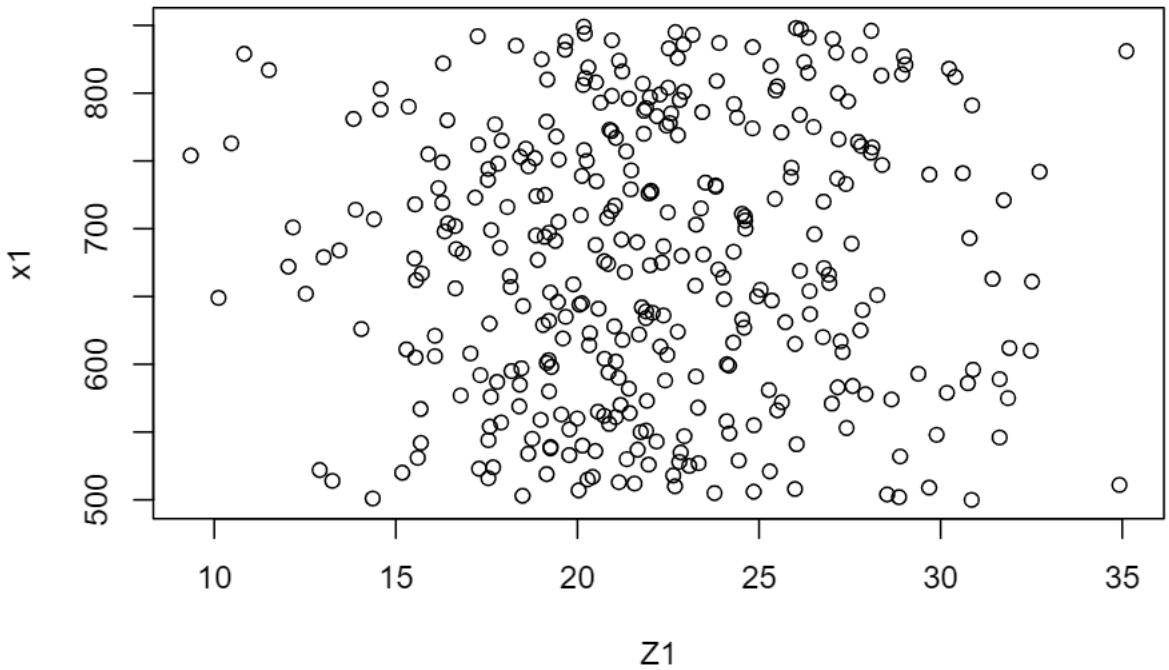
```r
# Graficar los puntos
plot(Z1)
```



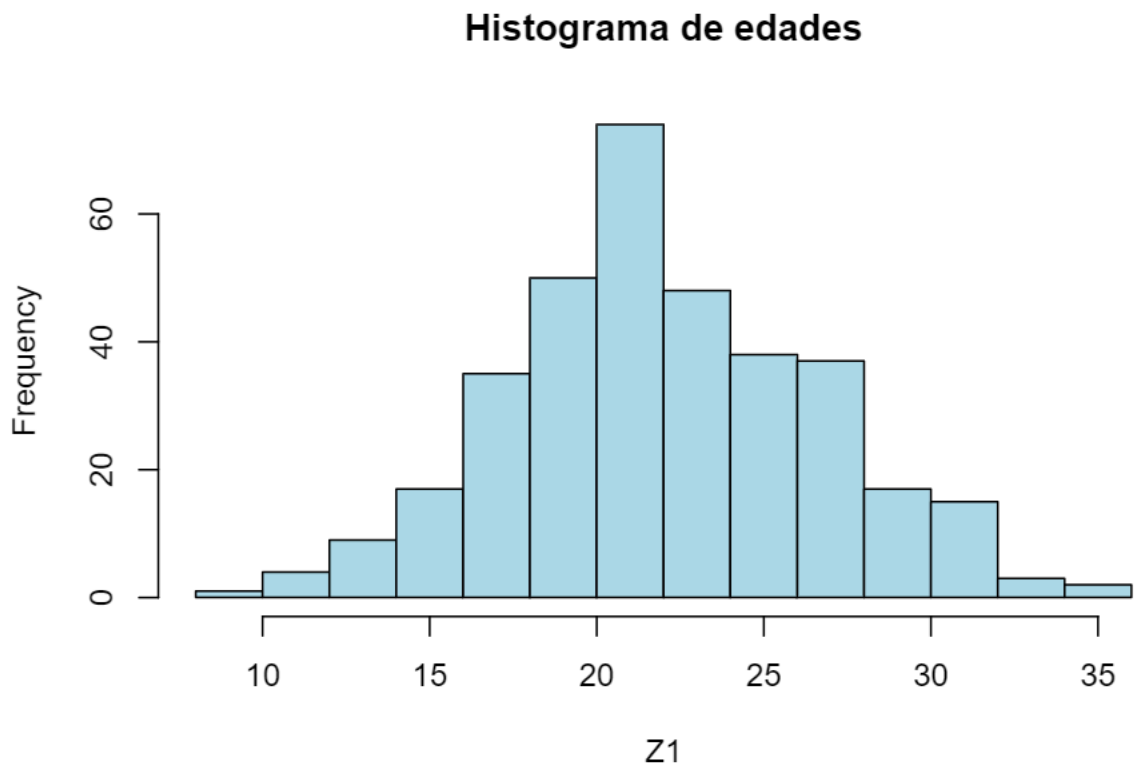

```r
# Obtener la longitud
w1 <- length(Z1)
print(w1)
```

```
## [1] 350
```

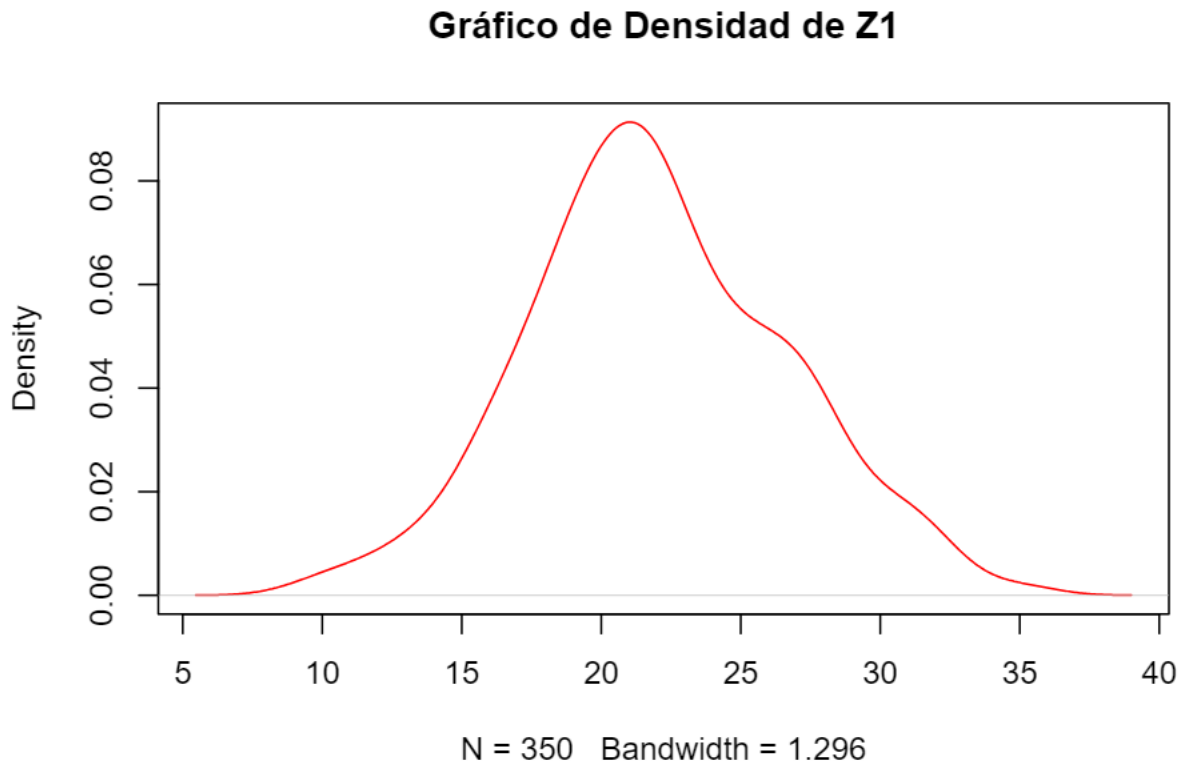
```r
# Crear una secuencia de la misma longitud para graficar (350 elementos)
x1 <- seq(500, length.out = 350)
plot(Z1, x1)
```




```r
# Creación de histograma
hist(Z1, main = "Histograma de edades", breaks = 10, col = "lightblue")
```




```r
# Cálculo y gráfico de densidad
densidad <- density(Z1)
plot(densidad, main = "Gráfico de Densidad de Z1", col = "red")
```



0.1 Ejercicio 1

DNI—910

```
#CONSIGNA: CREAR UN VECTOR QUE TENGA TODOS LOS NUMEROS ENTEROS DESDE EL 1 AL 910.
secuencia_dni<-(1:447)
print(secuencia_dni)
```

```
##   [1]   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18
##  [19]  19  20  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36
##  [37]  37  38  39  40  41  42  43  44  45  46  47  48  49  50  51  52  53  54
##  [55]  55  56  57  58  59  60  61  62  63  64  65  66  67  68  69  70  71  72
##  [73]  73  74  75  76  77  78  79  80  81  82  83  84  85  86  87  88  89  90
##  [91]  91  92  93  94  95  96  97  98  99 100 101 102 103 104 105 106 107 108
## [109] 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126
## [127] 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144
## [145] 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162
## [163] 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180
## [181] 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198
## [199] 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216
## [217] 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234
```

```
## [235] 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252
## [253] 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270
## [271] 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288
## [289] 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306
## [307] 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324
## [325] 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342
## [343] 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360
## [361] 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378
## [379] 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396
## [397] 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414
## [415] 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432
## [433] 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447
```

```
## Ejercicio 2
##Calcular la suma de todos los valores del vector secuencia_dni
```

```
Total<-0
valor_final<-length(secuencia_dni)
for (i in 1:valor_final)
Total<-Total+i
Total
```

```
## [1] 100128
```

0.2 Ejercicio 3

//Repetir ejercicio 2 pero usando Python
valor_final = len(secuencia_dni) total = 0
for i in range(1, valor_final + 1): total += i
print(total)

0.3 Ejercicio 4

//cuanto demora en correr el código del ejercicio 2

```
# Registramos el momento de inicio
inicio <- Sys.time()

Total <- 0
# Se corrige 'lenght' por 'length'
valor_final <- length(secuencia_dni)

for (i in 1:valor_final) {
  Total <- Total + i
}
```

```
# Registramos el momento de finalización
fin <- Sys.time()

# Calculamos la diferencia
duracion <- fin - inicio

# Mostramos los resultados
print(Total)
```

```
## [1] 100128
```

```
print(duracion)
```

```
## Time difference of 0.004403591 secs
```

0.4 Ejercicio 5

1 Implementación de una serie Fibonacci

La serie Fibonacci comienza con los números 0 y 1, a partir de estos cada uno de los siguientes términos es la suma de los dos anteriores.(0,1,1,2,3,5,8,13. . .) A continuación puede verse el código para implementar la serie:

```
F <- vector(length = 100)
A <- 0
B <-1
F[1] <-A
F[2] <-B
for (i in 3:100)
{F[i] <-(F[i-1]+F[i-2])}
head(F)
```

```
## [1] 0 1 1 2 3 5
```

Acknowledgements Please place your acknowledgments at the end of the paper, preceded by an unnumbered run-in heading (i.e. 3rd-level heading).

1. R Core Team: R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria (2023).
2. Wickham, H.: ggplot2: Elegant graphics for data analysis. Springer-Verlag New York (2016).
3. Xie, Y., Allaire, J.J., Grolemund, G.: R markdown: The definitive guide. Chapman; Hall/CRC, Boca Raton, Florida (2018).

4. Wickham, H., others: Welcome to the tidyverse. Journal of Open Source Software. 4, 1686 (2019).
5. RStudio Team: RStudio: Integrated development environment for r. (2020).